2	**(a)**	**(i)**	$v = u + at$	C1	
			$0 = 3.6 - 3.0t$		
			$t\ (= 3.6/3.0) = 1.2\,\text{s}$	A1	[2]
		(ii)	(distance to rest from P = $(3.6 \times 1.2)/2$ =) 2.2 (2.16) m **or** $[0 - (3.6)^2]/[2 \times (-3.0)] = 2.2\ (2.16)\,\text{m}$ **or** $3.6 \times 1.2 - ½ \times 3.0 \times (1.2)^2 = 2.2\ (2.16)\,\text{m}$ **or** $0 + ½ \times 3.0 \times (1.2)^2 = 2.2\ (2.16)\,\text{m}$	A1	[1]
	(b)		distance = 6.0 – 2.16 (= 3.84)	C1	
			$v^2 = u^2 + 2as = 2 \times 3.0 \times 3.84\ (= 23.04)$	M1	
			or		
			$x + 2 \times 2.16 = 6.0$ gives $x = 1.68$ (m)	(C1)	
			$v^2 = 3.6^2 + 2 \times 1.68 \times 3.0\ (= 23.04)$	(M1)	
			or correct method with intermediate time calculated ($t = 1.6\,\text{s}$ from Q to R)		
			$v = 4.8\,\text{m}\,\text{s}^{-1}$	A0	[2]

(c) straight line from $v = 3.6\ \mathrm{m\,s^{-1}}$ to $v = 0$ at $t = 1.2\ \mathrm{s}$ B1

straight line continues with the same gradient as v changes sign B1

straight line from $v = 0$ intercept to $v = -4.8\ \mathrm{m\,s^{-1}}$ B1 [3]

(d) difference in KE $= \frac{1}{2}m(v^2 - u^2)$

$= 0.5 \times 0.45\ (4.8^2 - 3.6^2)$ [= 5.184 – 2.916] C1

$= 2.3\ (2.27)\ \mathrm{J}$ A1 [2]